

I used to think..., Now I think

I used to think...	Now I think...	I still wonder...
I used to think that math fluency meant being fast at recalling facts, and that students who answered quickly were the strongest math learners.	Now I think that speed has very little to do with mathematical ability or true understanding. Boaler (2015) asserts "rapidity doesn't have a precise relation to intelligence. What is important is to deeply understand things and their relations to each other" (p. 5). The source youcubed (n.d.) shows at 0:23 that the image of a child slumped over her worksheet — disengaged and defeated — represents exactly what math should not look like for children.	I still wonder how I can help families unlearn the belief that being fast equals being smart in mathematics, especially when that belief has been reinforced for generations.
I used to think that timed tests were a neutral or even motivating practice for building math fact recall.	Now I think they are a significant source of harm for a large portion of students. Boaler (2015) asserts "for about one third of students the onset of timed testing is the beginning of math anxiety" (p. 3). This promotes that under stress felling were working memory becomes "blocked and students cannot access math facts they know" (p. 3). The source youcubed (n.d.) demonstrates at 0:44 that what we actually want for students looks entirely different. This look should be children leaning in together, using dice and markers, thinking actively and collaboratively rather than performing under pressure.	I still wonder how to advocate effectively for eliminating timed testing in schools where it is deeply entrenched, and what evidence or conversations are most likely to move administrators toward change.
I used to think that low-	Now I think the more accurate	I still wonder how to identify

achieving math students simply knew less content than their higher-achieving peers.	explanation is that struggling students have been directed toward memorization rather than flexible thinking from an early age. Boaler (2015) asserts on page 2 low achievers are often low achievers not because they know less but because they don't use numbers flexibly — they have been set on the wrong path, often from an early age, of trying to memorize methods instead of interacting with numbers flexibly. The source youcubed (n.d.) illustrates at 0:50 that the qualities we want math to embody — visual, collaborative, and engaging — are the very conditions that grow number sense in all learners.	students who have already internalized a memorization-only approach and what meaningful intervention looks like across different grade levels without making those students feel further behind.
• I used to think that flash cards were a harmless and helpful tool for building math fact fluency both at home and in the classroom.	• Now I think that traditional flash cards actually work against the development of real mathematical understanding. Boaler (2015) asserts in the section Math Cards on page 7 that flash cards usually include 2 unhelpful practices — memorization without understanding and time pressure. The source rather instead recommends a Math Cards activity where students match cards representing the same quantity through multiple representations such as area models, dot arrays, and number sentences, so that understanding drives the activity rather than speed. The source youcubed (n.d.) shows at 0:44 children engaged with physical tools and	• I still wonder how to design family communication around at-home math practice so that what families do at home supports rather than undermines the number sense approach being built in the classroom.

	each other in exactly this kind of meaningful, sense-making work.	
• I used to think that number sense was an innate quality — something students either naturally had or did not have.	• Now I think it is something that grows through rich, purposeful experiences with numbers and can be deliberately cultivated through thoughtful teaching. Boaler (2015) asserts on page 2 that number sense is the foundation for all higher-level mathematics and that students develop it when they work on rich mathematics problems rather than drilling isolated facts. The source youcubed (n.d.) demonstrates at 0:50 through classroom footage that when math is visual, collaborative, and engaging, students are naturally drawn into the kind of thinking that builds this foundation.	• I still wonder how to protect time in my future classroom for Number Talks, games like How Close to 100, and Snap It, and other number sense routines when scheduling pressures often push these rich activities aside in favor of more traditional drill-based practice.

References

Boaler, J. (2015, January 28). *Fluency without fear: Research evidence on the best ways to learn math facts*. youcubed at Stanford University.

youcubed. (n.d.). *Different experiences with math facts*. Stanford University.
<https://www.youcubed.org/resources/different-experiences-with-math-facts/>